Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 10 · day per timetable	10	1 period (60 min)	Dividing Polynomials	Algebra II	Topic 3 — Polynomial Functions · Lesson 3-4	Grade 10 — 10A and 10C

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · Whiteboards · Desmos (checking a remainder with p(a)) · Canva (final product) · Squared paper	Lesson presentation (15 slides) · "Choose Your Route" activity (3 pages) · Wayground kick-off

Competencies	Technology Embedded	Connection to student's real life
1- Dividing a polynomial by $x - a$ with synthetic division and reading the quotient and remainder. 2- Using $p(a)$ to find a remainder and to test a factor.	1- Quizizz live diagnostic with instant gap analysis. 2- Desmos: check that a zero of p gives a zero remainder.	1- A tank volume that factors into length, width and height. 2- Missing values in a model so that a division comes out exactly.

Values	Soft Skills	Hard Skills
1- Accuracy — the box number and every sign are written before they are used. 2- Precision — exact answers first.	1- Collaboration in assigned group roles. 2- Explaining in words why a zero remainder means a factor.	1- Proving $r = p(a)$ from $p(x) = (x - a)q(x) + r$. 2- Dividing with a placeholder zero.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Identify quotients and remainders upon the division of polynomials. 2- Apply the Remainder Theorem and the Factor Theorem. 3- Use synthetic division to divide a polynomial by a linear polynomial with leading coefficient 1. *Vocabulary: Divisor ; factor ; Polynomial ; long division ; proper rational function ; Quotient ; Remainder ; synthetic division.*	FIKR Phase 1 — Diagnose. 3-question Quizizz: evaluating a polynomial, factoring, listing coefficients. Results read as a gap map and used to route students. FIKR Phase 2 — Targeted Instruction. Long division; synthetic division; the Remainder and Factor Theorems; rewriting a rational expression. FIKR Phase 3 — Practice. Two identical "Give it a go" problems; then students choose a route and may switch. Compare routes and judge the claim. Pear Deck feedback. FIKR Phase 4 — Production. Qiddiya water-slide tank; AI as challenger only. FIKR Phase 5 — Proof of Learning. "Time to Check". No notes, no partner, no AI. FIKR Phase 6 — Smart Production. Make a cubic with a chosen factor, divide, swap and test; publish to the LMS portfolio.	Diagnostic quiz — gap map, not a grade. Whiteboard check (80% threshold) and "Give it a go" — self-marked. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. Final product graded on reasoning and clarity.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Two minutes of peer show-and-tell, then: "Which is harder — choosing the box number, or remembering the placeholder zero?"
Reflection for Improvement:	
Homework:	Upload the final product to the LMS portfolio. Pear Deck practice set (12 questions, choose your route). Schoology homework quiz for Lesson 3-4. Clinic group: bring your Time to Check paper.
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